

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: ITA 06

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO3: Bayes Theorem – Concept Learning – Maximum Likelihood – Minimum Description Length Principle – Bayes Optimal Classifier – Gibbs Algorithm – Naïve Bayes Classifier – Bayesian Belief Network – EM Algorithm – Probability Learning – Sample Complexity – Finite and Infinite Hypothesis Spaces – Mistake Bound Model, (BL 6)

Assessment Tool Weightage: 40%

QUESTIONS: 15

Total Marks:25

Annexure A

Experiments

Code of Conduct:

I P.Sai Ganesh Reddy(192525111) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

P.Sai Ganesh Reddy

Annexure A

1, Design and conduct an experiment to implement a Naïve Bayes classifier on a real-world dataset. Train the model using different feature sets and evaluate its performance using accuracy, precision, and recall. Analyze how assumptions of feature independence affect the results. Compare outcomes with and without Laplace smoothing, and interpret how probability estimation impacts classification performance.

Aim

To implement a Naïve Bayes classifier on a real-world dataset, evaluate its performance using accuracy, precision, and recall, and compare the effect of Laplace smoothing.

Algorithm

1. Import the required Python libraries.
2. Load the Iris dataset.
3. Split the dataset into training and testing sets.
4. Train a Gaussian Naïve Bayes classifier using the selected features.
5. Predict the classes of the test data.
6. Calculate accuracy, precision, and recall.
7. Train the classifier using different feature sets.
8. Compare the performance of the different feature sets.
9. Discuss the effect of the Naïve Bayes independence assumption.
10. Apply Laplace smoothing using a categorical dataset and compare the results.

Input

- Iris dataset containing sepal length, sepal width, petal length and petal width.
- Target classes: Setosa, Versicolor and Virginica.
- Feature sets:
 - All features
 - Sepal features
 - Petal features
- Categorical test sample: [2, 1].

Output

```
Feature Set: All Features
Accuracy : 0.9667
Precision: 0.9697
Recall   : 0.9667

Feature Set: Sepal Features
Accuracy : 0.7
Precision: 0.7003
Recall   : 0.7

Feature Set: Petal Features
Accuracy : 0.9667
Precision: 0.9697
Recall   : 0.9667

Without Laplace Smoothing:
[0]
With Laplace Smoothing:
[0]
```

Result

The Naïve Bayes classifier was successfully implemented. Different feature sets produced different classification performances. Laplace smoothing prevents zero probability estimates and can improve the reliability of classification when unseen feature combinations occur. The experiment also demonstrates the importance of the conditional independence assumption in Naïve Bayes.

2. Perform an experiment to estimate parameters of a probability distribution using Maximum Likelihood Estimation (MLE) and compare it with a Bayesian estimation approach. Use a dataset to compute parameters such as mean and variance, and observe how prior assumptions influence results. Analyze differences in estimation under small and large sample sizes, and interpret the impact on model reliability.

Aim

To estimate the parameters of a probability distribution using Maximum Likelihood Estimation (MLE) and Bayesian estimation, and study the effect of prior assumptions and sample size.

Algorithm

1. Generate or load normally distributed data.
2. Calculate the sample mean and variance using MLE.
3. Define prior parameters for Bayesian estimation.
4. Calculate the posterior mean and variance.
5. Repeat the estimation for a small sample.
6. Repeat the estimation for a large sample.
7. Compare MLE and Bayesian estimates.
8. Analyze the influence of prior information.
9. Observe how increasing the sample size affects reliability.

Input

- Normally distributed sample data.
- Population mean = 50.
- Population standard deviation = 10.
- Small sample size = 10.
- Large sample size = 100.
- Prior mean = 45.
- Prior variance = 25.

Output

```
SMALL SAMPLE
MLE Mean      : 54.4806
MLE Variance  : 47.0467

LARGE SAMPLE
MLE Mean      : 48.9615
MLE Variance  : 81.6522

BAYESIAN ESTIMATION - SMALL SAMPLE
Posterior Mean      : 51.7719
Posterior Variance  : 7.1429

BAYESIAN ESTIMATION - LARGE SAMPLE
Posterior Mean      : 48.8092
Posterior Variance  : 0.9615
```

Result

MLE and Bayesian parameter estimation were successfully implemented. For small samples, prior assumptions have a greater influence on Bayesian estimates. As the sample size increases, the observed data becomes more dominant and the Bayesian estimate approaches the MLE.

3. Implement the Expectation-Maximization (EM) algorithm for a mixture model and conduct experiments to study its convergence behavior. Vary initialization parameters and number of clusters, and observe their effect on convergence speed and final likelihood values. Record and analyze cases where the algorithm converges to local optima. Suggest strategies to improve convergence stability and performance.

Aim

To implement the Expectation-Maximization (EM) algorithm for a Gaussian mixture model and study the effect of initialization and number of clusters on convergence and likelihood.

Algorithm

1. Generate a dataset containing samples from multiple Gaussian distributions.
2. Select the number of clusters.
3. Initialize the Gaussian mixture model parameters.
4. Perform the Expectation step to calculate the probability that each sample belongs to

each cluster.

5. Perform the Maximization step to update the means, variances and mixture weights.
6. Calculate the log-likelihood.
7. Repeat the E-step and M-step until convergence.
8. Change the initialization parameters.
9. Change the number of clusters.
10. Compare convergence speed and final likelihood values.
11. Identify possible local optima.

Input

- Generated dataset with 300 samples.
- Actual number of clusters = 3.
- Cluster configurations tested:
 - 2 clusters
 - 3 clusters
 - 3 clusters with different initialization
 - 4 clusters

Output

```
2 Clusters
Iterations : 2
Log Likelihood: -4.5325
Means:
[[ 1.05693535  5.52708203]
 [-6.88387179 -6.98398415]]

3 Clusters
Iterations : 2
Log Likelihood: -3.8686
Means:
[[-2.63323268  9.04356978]
 [-6.88387179 -6.98398415]
 [ 4.74710337  2.01059427]]

3 Clusters - Different Initialization
Iterations : 2
Log Likelihood: -3.8686
Means:
[[-6.88387179 -6.98398415]
 [-2.63323268  9.04356978]
 [ 4.74710337  2.01059427]]

4 Clusters
Iterations : 6
Log Likelihood: -3.8584
Means:
[[-2.63323268  9.04356978]
 [-6.88387179 -6.98398415]
 [ 5.24840118  1.78154552]
 [ 4.1465995   2.28497142]]
```

Result

The EM algorithm was successfully implemented using a Gaussian mixture model. The number of clusters and initialization affect convergence speed and the final likelihood. Different initializations can lead to different local optima. Multiple initializations and suitable selection of the number of clusters can improve convergence stability.

4. Construct a Bayesian Belief Network for a given problem domain and perform inference under different evidence conditions. Conduct experiments by modifying conditional probabilities and observing changes in posterior probabilities. Analyze how dependencies between variables influence the results. Evaluate the robustness of the model and interpret how uncertainty is handled in probabilistic reasoning.

Aim

To construct a Bayesian Belief Network and perform probabilistic inference under different evidence conditions by modifying conditional probabilities.

Bayesian Network

The following simple network is considered:

Cloudy

/ \

Rain Sprinkler

\ /

Wet Grass

Algorithm

1. Define the random variables in the Bayesian network.
2. Define prior probabilities for the root node.
3. Define conditional probability tables for dependent variables.
4. Enter evidence for selected variables.
5. Calculate the posterior probability of the required variable.
6. Modify conditional probabilities.
7. Repeat the inference.
8. Compare the posterior probabilities.
9. Analyze how dependencies affect the final result.

Input

Evidence conditions:

1. Rain = True
2. Sprinkler = True

Conditional probabilities are specified in the Bayesian network.

Output

Bayesian Belief Network

$$P(\text{Rain}) = 0.3$$

$$P(\text{No Rain}) = 0.7$$

$$P(\text{Wet Grass} \mid \text{Rain}) = 0.8$$

$$P(\text{Wet Grass} \mid \text{No Rain}) = 0.1$$

Evidence: Wet Grass = True

$$P(\text{Rain} \mid \text{Wet Grass}) = 0.7742$$

$$P(\text{No Rain} \mid \text{Wet Grass}) = 0.2258$$

Result

A Bayesian Belief Network was successfully constructed and inference was performed under different evidence conditions. The experiment shows that changing evidence or conditional probabilities changes posterior probabilities. Dependencies between variables therefore play an important role in probabilistic reasoning and uncertainty handling.

5. Design and conduct an experiment to implement a **Decision Tree classifier** on a real-world dataset. Train the model using different splitting criteria such as entropy and Gini index, and evaluate performance using accuracy and F1-score. Analyze the impact of tree depth and pruning on overfitting, and compare results across different configurations.

Aim

To implement a Decision Tree classifier using different splitting criteria such as entropy and Gini index, and evaluate the classifier using accuracy and F1-score.

Algorithm

1. Import the required libraries.
2. Load the Iris dataset.
3. Separate features and target variables.
4. Split the dataset into training and testing sets.
5. Create a Decision Tree using the Gini index.
6. Train the model.
7. Predict the test data.
8. Calculate accuracy and F1-score.
9. Repeat the process using entropy.
10. Compare the results and analyze the effect of tree depth.

Input

Iris dataset with:

Features:

Sepal Length

Sepal Width

Petal Length

Petal Width

Target:

Setosa, Versicolor, Virginica

Maximum Tree Depth = 3

Output

```

Gini Index
Accuracy : 1.0
F1 Score : 1.0

Entropy
Accuracy : 0.9777777777777777
F1 Score : 0.9777448559670782

```

Result

Thus, the Decision Tree classifier was successfully implemented using both Gini index and entropy, and the classification performance was evaluated using accuracy and F1-score.

RUBRICS FOR CALCULATIO

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	15	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand